

1. Clocking methodology is a synchronization mechanism in processor where all operations are coordinated using a clock signal. Each clock pulse triggers data movement b/w registers and execution is functional units ensuring orderly and timed execution of instructions.

2. A clock cycle is the smallest unit of time in processor operation. Each instruction is broken into steps executed in one or more clock cycles. Processor speed is inversely proportional to clock cycle time.

3. Amdahl's law: defines the theoretical speedup of a system when a part of it is improved.

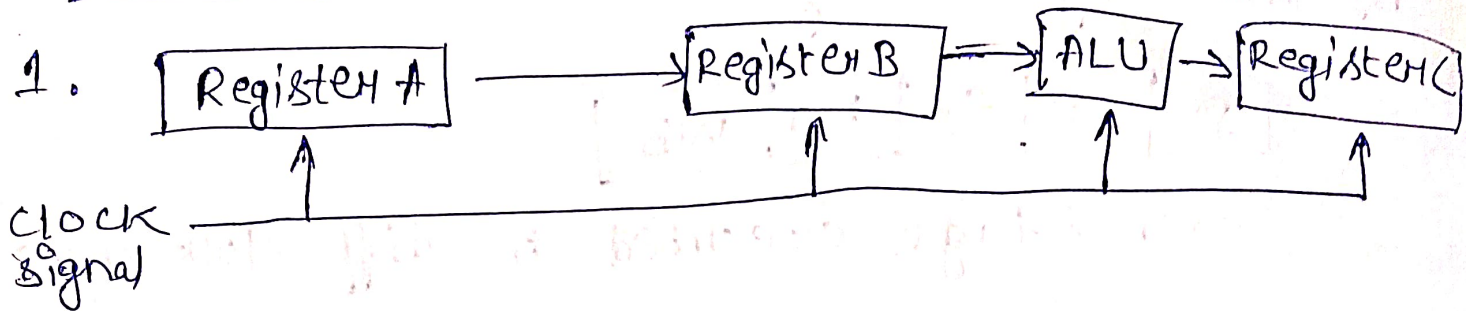
$$\text{speedup} = \frac{1}{(1-P) + (P/S)}$$

P = parallel portion of the program
S = speedup of that portion.

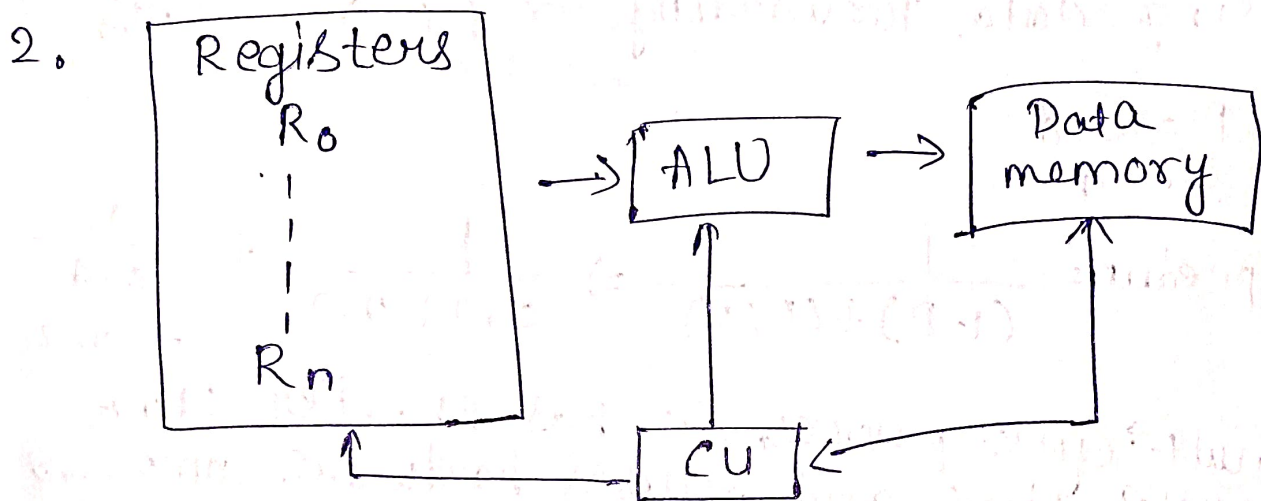
4. The datapath consists of hardware components responsible for processing data such as ALU, registers, buses and multiplexers. It is the part of the CPU that actually executes the instructions.

6. A single cycle processor executes an entire instruction in one clock cycle.
7. A multi cycle processor breaks the instruction execution into multiple steps. Each step is executed in a separate clock cycle.
8. Instruction Pipelining is a technique in which the execution of multiple instructions is overlapped by dividing the instruction cycle into a no. of stages.
9. ILP is the ability of a processor to execute multiple instruction by exploiting independent instructions in the program.
10. Data hazards occur in pipelining when an instruction depends on the result of previous instruction which has not completed its execution.
11. control hazard occur due to branch instruction which change the normal flow of the program.
12. Pipeline hazards can be mitigated using
 - ↳ Forwarding
 - ↳ Pipeline stalls
 - ↳ Branch Prediction
 - ↳ Instruction re-ordering

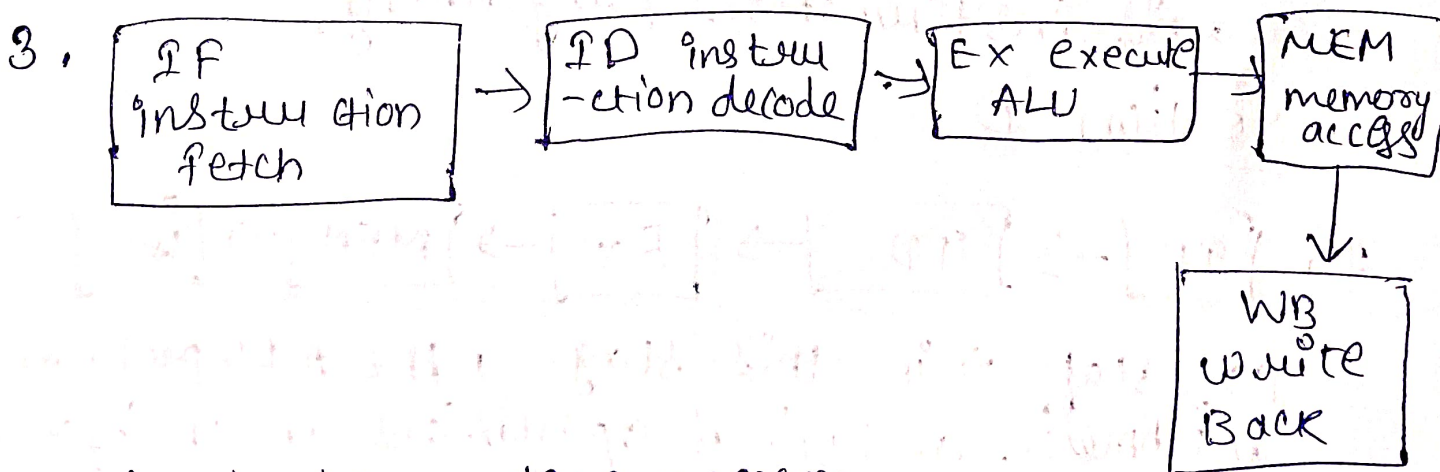
Section-B



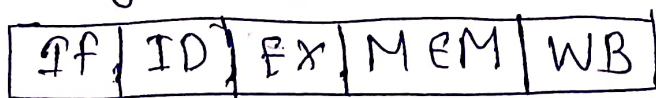
The clock signal synchronizes the operations of all registers and functional units.



Data flows from registers to ALU for processing, then to memory or back to register.

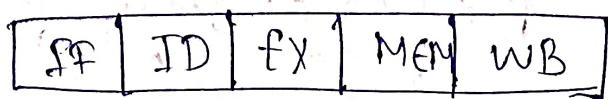


4. a) Single cycle processor



entire instruction executed in 1 long clock cycle.

b) multi cycle processor



each stage executed in diff clock cycles.

Section - C

1. Data hazards will occur. They can be solved using data forwarding or pipeline stalls.

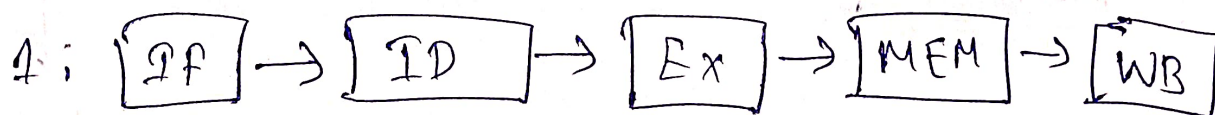
$$2. \quad P = 0.8 \\ S = 4$$

$$\text{speedup} = \frac{1}{(1-P) + (P/S)} \Rightarrow \frac{1}{0.2 + 0.2} = \frac{1}{0.4} = \frac{1}{0.2} = 5$$

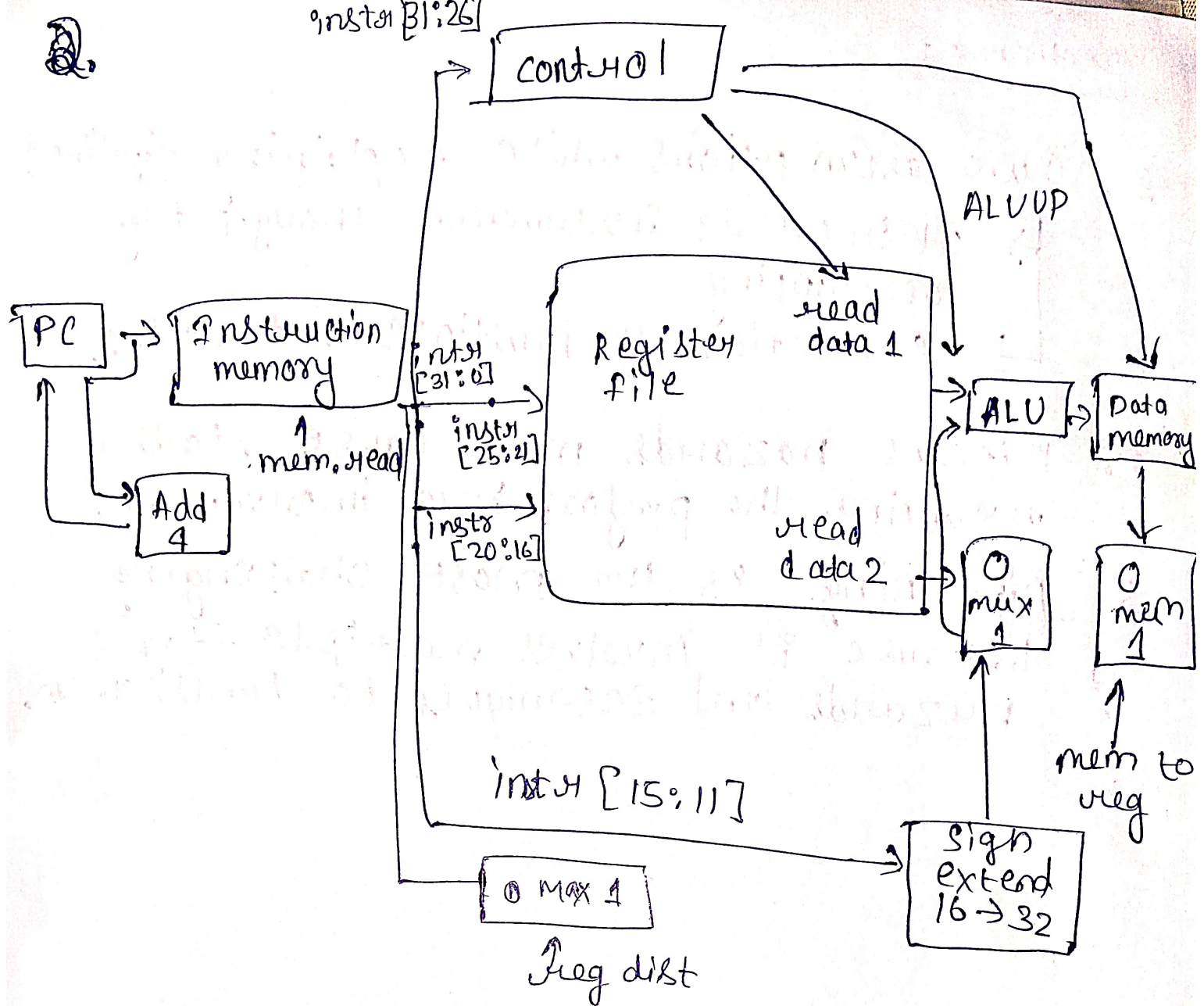
3. multicyle processor has shorter clock cycle time and reuses hardware component for diff stages.

4. control hazards occur. It can be reduced by using branch prediction techniques and instruction prefetching.

Section - D



EX stage → in this stage, the ALU perform arithmetic or logical operations on the operands. It may also compute branch target address.



it reduces hardware cost and complexity since we need only one memory interface.

3. Example

ADD R_1, R_2, R_3 ; $R_1 = R_2 + R_3$

SUB R_4, R_1, R_5 ; $R_4 = R_1 - R_5$

Prob: Sub needs R_1 before ADD writes it back.

Soln: Use data forwarding from EX/MEM stage to EX stage or insert one stall.

Section - F

1. Two assumptions while explaining pipeline
 - ↳ It increase instruction throughput by overlapping
 - ↳ execution of multiple instructions.
2. Pipeline hazards may cause stalls, reducing the performance improvement.
3. Pipelining is the most challenging because it involves multiple stages hazards and techniques to handle them.